## NAME

**mbswath2las** − exports swath bathymetry soundings to LAS 1.4 point cloud files.

## VERSION

Version 5.0

## SYNOPSIS

**mbswath2las**
[
**−−use-amplitude** {**−A**}
**−−begin-time**=*yr/mo/da/hr/mn/sc* {**−B***yr/mo/da/hr/mn/sc*}
**−−end-time**=*yr/mo/da/hr/mn/sc* {**−E***yr/mo/da/hr/mn/sc*}
**−−format**=*format* {**−F***format*}
**−−help** {**−H**}
**−−input**=*file* {**−I***file*}
**−−projection**=*projection* {**−J***projection*}
**−−lonflip**=*lonflip* {**−L***lonflip*}
**−−output**=*file* {**−O***file*}
**−−write-prj** {**−P**}
**−−bounds**=*west/east/south/north* {**−R***west/east/south/north*}
**−−speed-minimum**=*speed* {**−S***speed*}
**−−time-gap**=*timegap* {**−T***timegap*}
**−−verbose** {**−V**}
]

## DESCRIPTION

**mbswath2las** reads swath bathymetry data in any format supported by **MBIO** and writes the individual soundings as points in LAS format files, one point per valid (unflagged) beam. This allows swath bathymetry to be loaded directly into LAS/LIDAR-oriented point cloud viewers, GIS packages, and processing pipelines (e.g. CloudCompare, QGIS, PDAL) that do not otherwise read **MBIO** formats.

For each beam, the sounding position is computed from the ship navigation, heading, and the beam's acrosstrack/alongtrack offsets, following the same geometry used by **mblist**. By default the position is written as geographic longitude/latitude in decimal degrees (WGS84). If the **−J** option is used to specify a projected coordinate system, the position is instead written as easting/northing in meters. In both cases the LAS elevation (Z) value is the negative of the sounding depth, since LAS elevations are positive up while **MBIO** bathymetry values are positive down. Beams flagged bad or otherwise unusable are skipped and do not appear in the output.

By default **mbswath2las** reads each input swath file's faster ".fbt" derivative when one is available, since this program otherwise only needs bathymetry; ".fbt" files do not carry amplitude, so LAS intensity is left 0 unless **−−use-amplitude** is given, in which case the original swath file is read instead (more slowly) so that real per-beam amplitude is available. When populated, the intensity value is the input data's per-beam sonar amplitude linearly rescaled from the minimum/maximum amplitude found across the entire input (from the ".inf" files, the same source used for **−−projection**'s automatic bounds, see below) into the unsigned 16-bit (0-65535) range required by the LAS format; this preserves relative amplitude regardless of whether the input format reports amplitude as signed dB (as Kongsberg and other sonars commonly do) or as raw positive counts, but it is not a radiometrically calibrated reflectivity value. The LAS point's GPS time field is set to the sounding's Unix epoch timestamp (see **BUGS** below).

**mbswath2las** automatically generates any missing or out-of-date ".inf", ".fbt", and ".fnv" ancillary files for its input (the same files **mbdatalist**(1)'s **−N** option produces), both to support the ".fbt" read path above and to determine survey bounds for automatic projection selection (see **−−projection** below).

Some GIS software (notably ArcGIS) looks for the coordinate system of a point cloud or other geospatial file in an accompanying ".prj" sidecar file rather than (or in addition to) reading it from the file itself. If the **−−write-prj** option is given, **mbswath2las** writes such a ".prj" file next to each LAS output file, containing the same coordinate system as the LAS file's own WKT VLR (see **LAS OUTPUT FORMAT** below), but in the older, Esri-specific WKT1 dialect that ".prj" files conventionally use rather than the WKT2 used in the VLR.

**mbswath2las** operates on a single input swath file or on a datalist referencing many swath files (see **mbdatalist**(1)). By default, a separate LAS output file is created for each input swath file, named by appending ".las" to the input file name. If the **−O** option is used to specify an output file name, all of the soundings from every input file are instead collected into that single LAS file.

**mbswath2las** writes standard LAS 1.4 files (public header block, an "OGC Coordinate System WKT" variable length record describing the sounding coordinate reference system, and point data record format 1) using its own internal writer; no third-party LAS/LIDAR library (e.g. libLAS, LASzip, PDAL) is required to build or run this program, though the coordinate system description is obtained from the **PROJ** library. See **LAS OUTPUT FORMAT** below for details.

## MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian dos Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

## OPTIONS

**−−use-amplitude {−A}**

Populates the LAS intensity field from each sounding's per-beam sonar amplitude. This forces **mbswath2las** to read the original swath file(s) rather than substituting the faster ".fbt" derivative, since ".fbt" files do not carry amplitude. Default: intensity is left 0, and the ".fbt" file is used in place of the original swath file when one is available.

**−−begin-time=***yr/mo/da/hr/mn/sc* **{−B***yr/mo/da/hr/mn/sc***}**

This option sets the starting time for data allowed in the input data. The **−E** option sets the ending time for data. If the starting time is before the ending time, then any data with a time stamp before the starting time or after the ending time is ignored. Default: *yr/mo/da/hr/mn/sc* = 1962/2/21/10/30/0, i.e. of no effect.

**−−end-time=***yr/mo/da/hr/mn/sc* **{−E***yr/mo/da/hr/mn/sc***}**

This option sets the ending time for data allowed in the input data. The **−−begin-time** option sets the starting time for data. Default: *yr/mo/da/hr/mn/sc* = 2062/2/21/10/30/0, i.e. of no effect.

**−−format=***format* **{−F***format***}**

Sets the data format of the input swath file(s). If *format* < 0, then the file specified with the **−I** option is a datalist referencing multiple swath data files and their formats (see **mbdatalist**(1)). Default: the format is guessed from the input file's suffix, as set by **mbdefaults**(1).

**−−help {−H}**

This "help" flag causes the program to print out a description of its operation and then exit immediately.

**−−input=***file* **{−I***file***}**

Sets the input file path. If *format* > 0, *file* is a single swath data file. If *format* < 0, *file* is an ASCII datalist naming one or more swath data files, each followed by its **MBIO** format identifier, e.g.:

datafile1 88
datafile2 88

Default: *file* = "datalist.mb−1".

**−−projection=***projection* **{−J***projection***}**

Specifies a projected coordinate system (PCS) to be used for the output sounding positions in place of the default geographic (longitude/latitude, WGS84) coordinates. When a PCS is specified, positions are written to the LAS file as eastings and northings in meters. The *projection* argument may be a PCS identifier as accepted by **mblist**(1) and **mbgrid**(1); simply **U** (or **UTM**) to use Universal Transverse Mercator in whatever zone contains the center of the input data's geographic bounds; simply **L** (or **LTM**) for a local-origin Transverse Mercator projection centered on that same bounds center, with unit scale factor and zero false easting/northing (so that north lies along the LAS Y axis); **LTM***lon0*/*lat0* for a local-origin Transverse Mercator projection centered at an explicit *lon0*/*lat0* instead; or any coordinate system string understood directly by the **PROJ** library (e.g. an EPSG code, or a full **+proj=**... definition). The bare **U**/**UTM** and **L**/**LTM** forms determine the input data's geographic bounds the same way **mbgrid**(1) and **mbmosaic**(1) do: from each input file's ".inf" sidecar (generating it first if missing or out of date, see above), not by reading any sounding data. Whichever coordinate system is selected is recorded in the output LAS file's "OGC Coordinate System WKT" variable length record (see **LAS OUTPUT FORMAT** below), so downstream software can recover it automatically.

**−−lonflip=***lonflip* **{−L***lonflip***}**

Sets the range of the longitude values used. If *lonflip*=−1 then the longitude values will be in the range from −360 to 0 degrees. If *lonflip*=0 then the longitude values will be in the range from −180 to 180 degrees. If *lonflip*=1 then the longitude values will be in the range from 0 to 360 degrees. Default: *lonflip* is set by **mbdefaults**(1).

**−−output=***file* **{−O***file***}**

Sets the output LAS file path. If **−O** is given, all soundings from every input swath file are written to this single LAS file. If **−O** is not given, one LAS output file is created per input swath file, named by appending ".las" to the input swath file name (e.g. *survey.mb59* produces *survey.mb59.las*).

**−−write-prj {−P}**

Writes an Esri-style ".prj" sidecar file next to each LAS output file, containing the coordinate system used, for GIS software that looks for one there instead of (or in addition to) reading it from the LAS file itself. Named by replacing the LAS output file's ".las" suffix with ".prj" (e.g. *survey.las* produces *survey.prj*). Default: no ".prj" file is written.

**−−bounds=***west/east/south/north* **{−R***west/east/south/north***}**

Sets the longitude and latitude bounds within which swath sonar data will be read. Only the data which lies within these bounds will be output to the LAS file(s). Default: *west/east/south/north* = −360/360/−90/90, i.e. of no effect.

**−−speed-minimum=***speed* **{−S***speed***}**

Sets the minimum speed in km/hr (5.5 kts ˜ 10 km/hr) allowed in the input data; pings associated with a smaller ship speed will not be output to the LAS file(s). Default: *speed* = 0, i.e. of no effect.

**−−time-gap=***timegap* **{−T***timegap***}**

Sets the maximum time gap in minutes between adjacent pings allowed before the data is considered to have a gap. Time gaps do not stop **mbswath2las** from processing an input file; the option only affects how gaps are reported when **−−verbose** is used. Default: *timegap* = 1.

**−−verbose {−V}**

This option increases the verbosity of **mbswath2las**, causing it to print out the coordinate system actually used -- the projection identifier when **−J** is given (useful with the bare **U**/**UTM** and **L**/**LTM** forms, since the specific zone or origin selected is otherwise not shown), or a note that plain geographic (WGS84) coordinates are being used otherwise -- the name of each LAS output file as it is opened, any comments encountered in the input data, and a final count of the total number of LAS points written. This option can be repeated (e.g. **−VV**) to obtain more detailed **MBIO** debugging output.

## LAS OUTPUT FORMAT

**mbswath2las** writes standard LAS 1.4 files with the following properties:

- A 375-byte LAS 1.4 public header block, with the Global Encoding WKT bit (bit 4) set to indicate that the coordinate reference system is described by WKT rather than the older GeoTIFF key mechanism used by LAS 1.2/1.3. LAS 1.4 (rather than the 1.2 header used by earlier versions of this program) is used specifically so that this WKT-based CRS description can be written; LAS 1.5 goes further and requires WKT while dropping GeoTIFF keys entirely, but also drops point data record formats 0−5, which is why this program does not target it.
- A single "OGC Coordinate System WKT" variable length record (User ID "LASF_Projection", Record ID 2112) immediately following the header, containing a null-terminated WKT2 description of the sounding coordinate system: plain geographic WGS84 (EPSG:4326) by default, or the actual projected coordinate system requested with **−J** (including a fully custom definition, e.g. a local-origin Transverse Mercator specified as **−J***LTMlon0/lat0*, or as a raw PROJ string). This WKT is obtained directly from **PROJ**, so it always matches the actual projection used, whatever it is.
- Point data record format 1 (X, Y, Z, intensity, return/classification flags, scan angle, user data, point source ID, and GPS time), 28 bytes per point.
- X/Y scale factor: 1e−7 degree (geographic mode) or 0.01 m (projected mode). Z scale factor: 0.001 m. No coordinate offset is applied; these scale factors comfortably span a full survey within the signed 32−bit integer range used for LAS point coordinates.
- All multi−byte fields are stored little−endian per the LAS specification. On a big−endian host, **mbswath2las** byte−swaps every field on write so the output file is always correct regardless of build platform.

## EXAMPLES

Convert a single swath file to a LAS file, using the default output name (adds a ".las" suffix):

```
mbswath2las −I 0002_20180911_020351_EX1811_MB.mb88 −F 88
```

which produces *0002_20180911_020351_EX1811_MB.mb88.las*.

Convert every file referenced by a datalist to its own LAS file, printing progress and a final point count:

```
mbswath2las −−input=datalist.mb−1 −−verbose
```

Collect an entire datalist's soundings into a single combined LAS file, projected to UTM instead of geographic coordinates:

```
mbswath2las −−input=datalist.mb−1 −−projection=UTM \
    −−output=survey.las
```

Same as above, but centered on a local Transverse Mercator origin instead of UTM, and with intensity populated from the original swath files' amplitude data (reading the ".fbt" files would otherwise be faster but leaves intensity at 0):

```
mbswath2las −−input=datalist.mb−1 −−projection=LTM \
    −−use-amplitude −−output=survey.las
```

Same as the UTM example above, but also write a ".prj" sidecar file (*survey.prj*) for GIS software that expects one:

```
mbswath2las −−input=datalist.mb−1 −−projection=UTM \
    −−write-prj −−output=survey.las
```

## SEE ALSO

**mbsystem**(1), **mbswath**(1), **mblist**(1), **mbcopy**(1), **mbinfo**(1), **mbdatalist**(1), **mbgrid**(1), **mbmosaic**(1)

## BUGS

The LAS point GPS time field is set to the raw Unix epoch timestamp of each sounding rather than the leap-second-adjusted GPS time nominally expected by the LAS format; this is sufficient for ordering and lookup of points by time but is not standards-compliant GPS time.

When **−−use-amplitude** is given, sounding intensity is rescaled linearly from the minimum/maximum amplitude found across the entire input (read from the ".inf" files) into the unsigned 16-bit (0-65535) range required by LAS, so it is not a radiometrically calibrated reflectivity value. If the ".inf" files report a degenerate range (maximum not greater than minimum, e.g. an input format with no amplitude data at all), intensity is left 0 throughout rather than divided by zero.